

Lab Practice Problem 1

Filename: lab1.c

In this activity, you will complete a partially implemented C program that reads song records from a file, stores them in a playlist, prints the playlist, and identifies the longest song. This exercise is intended to reinforce your skills in file input, string handling, arrays, structures, and function implementation.

Try It Out

Write a program that reads up to 5 songs from an input file, where each song record contains a song ID, a title, and a length in seconds. Store each song into a `Playlist` structure, then print all songs in the playlist. Finally, find the longest song and print its information.

Assumptions

You may assume each song title contains no whitespace (use `_` as a separator) and is at most 100 characters long. You may also assume the input file contains valid integers for the song ID and length. Further, no two songs will have the same length.

Your task is to complete the given C program. You can access the starter code [here](#). The following function prototypes describe the functions you will implement:

```
void playlist_init(Playlist *list, int id);
```

Initializes a playlist by setting its `list_id` to `id` and setting count to 0.

```
void playlist_print(Playlist *list);
```

Prints the playlist ID on the first line using the format `Playlist ID: <id>` (see Sample Output). Then prints each song currently stored in the playlist, one per line, by calling `song_print()` for each song.

```
void playlist_add_song(Playlist *list, int s_id, char *s_title, int s_len);
```

Adds a new song to the playlist by storing the given ID, title, and length into the next available position in songs. Increments the playlist's count after inserting the song.

```
Song *playlist_find_longest_song(Playlist *list);
```

Returns a pointer to the song with the greatest length in the playlist. If the playlist is empty, return `NULL`.

```
void song_print(Song *s);
```

Prints a song's information in single line using the following format: `title - length`

Important You must not modify any other parts of the program. Your only task is to complete the definitions of the five functions listed above. Do not add new helper functions.

Input

The program reads input from a file named `songs.txt`, which contains song records stored across multiple lines. Each song is described using three consecutive lines: an integer representing the song ID, a title string containing no whitespace characters, and an integer representing the song's length in seconds. The program processes up to 5 songs from the file in the order they appear.

Output

The program first prints the playlist ID, followed by each song stored in the playlist on its own line. Afterward, it prints the label Longest Song: and displays the title of the song with the greatest length.

`songs.txt`

```
1
Shape_of_You
233
2
Blinding_Lights
200
3
Bohemian_Rhapsody
354
4
Bad_Guy
194
5
Supercalifragilisticexpialidocious
196
```

Sample Output

```
Playlist ID: 101
Shape_of_You - 233
Blinding_Lights - 200
Bohemian_Rhapsody - 354
Bad_Guy - 194
Supercalifragilisticexpialidocious - 196
Longest Song: Bohemian_Rhapsody
```

Guide Questions

1. What happens if the input file contains information for more than five songs?
2. What occurs if `playlist_add_song()` is called more than five times, exceeding the playlist's maximum capacity?
3. If the assumption that all songs have unique lengths is removed, how does the program behave when two or more songs have the same length? Which song title will be printed, and why?

Sample Solution

The source code can be accessed [here](#).